

STANDARD 2

Stage 6

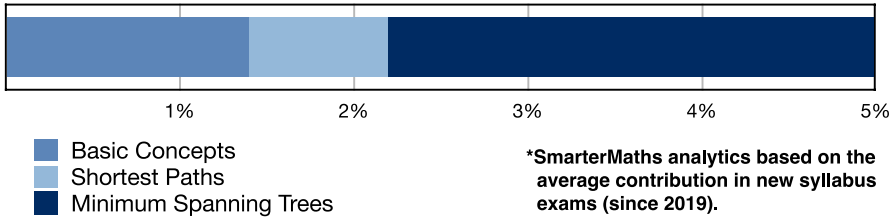
Networks (Std 2), N2 Network Concepts (Y12)

Shortest Paths

Teacher: Hailey Summer

Exam Equivalent Time: 60 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History N2 Network Concepts



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-N2 Network Concepts* has contributed 5.0% to new syllabus exams since its introduction in 2019.
- We have split this area into three categories for the purposes of analysis: 1-*Basic Concepts* (1.4%), 2-*Shortest Paths* (0.8%) and 3-*Minimum Spanning Trees* (2.8%).
- This analysis will look at the smallest sub-topic, *Shortest Paths* (0.8%).

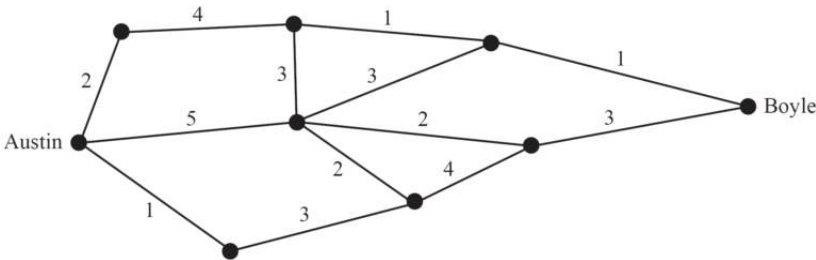
ANALYSIS - What to Expect and Common pitfalls

- *Shortest Paths* is typically examined at low to mid-band difficulty level and has been tested in 2023, 2021 and 2019 in new syllabus exams, usually as a part of a minimum spanning tree question.
- Implementing *Dijkstra's algorithm* to solve questions in this topic has been a focus. This solution method is provided in detail in most worked solutions, reflecting its importance in NESA's *Networks Topic Guide*.
- Questions can require students to find a "shortest distance" by providing either a network diagram or a table, both of which are covered in the database.

Questions

1. Networks, STD2 N2 2009 FUR1 2 MC

The network shows the distances, in kilometres, along roads that connect the cities of Austin and Boyle.

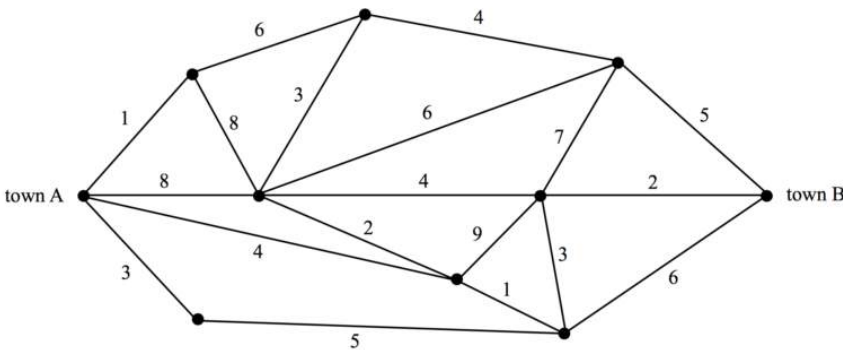


The shortest distance, in kilometres, from Austin to Boyle is

- A. 7
- B. 8
- C. 9
- D. 10

2. Networks, STD2 N2 2007 FUR1 4 MC

The following network shows the distances, in kilometres, along a series of roads that connect town **A** to town **B**.

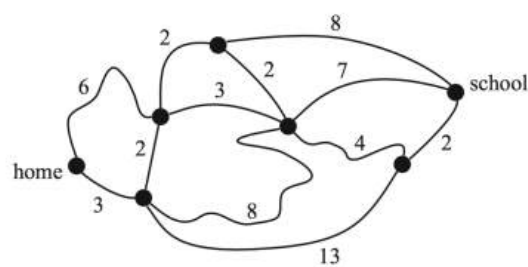


The shortest distance, in kilometres, to travel from town **A** to town **B** is

- A. 9
- B. 10
- C. 11
- D. 12

3. Networks, STD2 N2 2014 FUR1 3 MC

The diagram below shows the network of roads that Stephanie can use to travel between home and school.



The numbers on the roads show the time, in minutes, that it takes her to ride a bicycle along each road. Using this network of roads, the shortest time that it will take Stephanie to ride her bicycle from home to school is

- A. 12 minutes
- B. 13 minutes
- C. 14 minutes
- D. 15 minutes

4. Networks, STD2 N2 SM-Bank 25 MC

In a town, there are four cafes **W**, **X**, **Y** and **Z**. The table shows the distances, in metres, of paved footpath connecting the cafes.

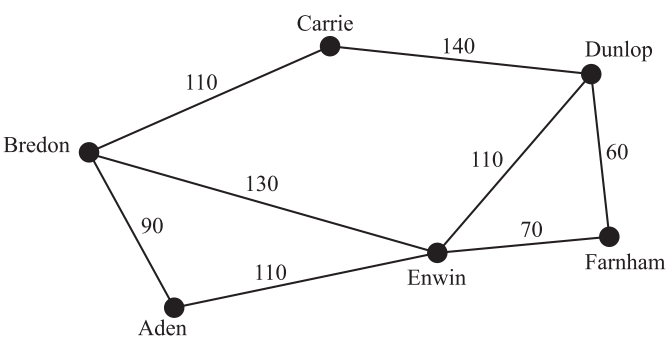
	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>W</i>	—	150	—	120
<i>X</i>	150	—	140	70
<i>Y</i>	—	140	—	100
<i>Z</i>	120	70	100	—

A coffee supplier needs to visit each cafe. What is the shortest distance she needs to walk along the paved footpath if she starts at cafe **W**?

- A. 260 m
- B. 320 m
- C. 330 m
- D. 360 m

5. Networks, STD2 N2 2011 FUR2 1

Aden, Bredon, Carrie, Dunlop, Enwin and Farnham are six towns. The network shows the road connections and distances between these towns in kilometres.



- a. In kilometres, what is the shortest distance between Farnham and Carrie? (1 mark)
- b. How many different ways are there to travel from Farnham to Carrie without passing through any town more than once? (1 mark)

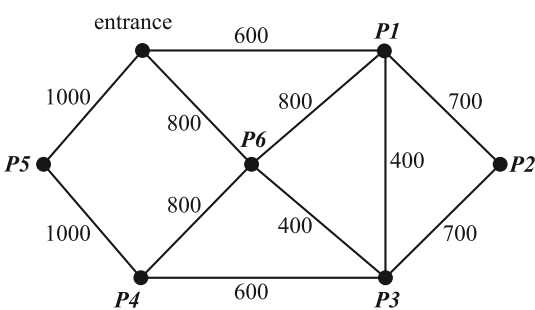
An engineer plans to inspect all of the roads in this network. He will start at Dunlop and inspect each road only once.

- c. At which town will the inspection finish? (1 mark)

6. Networks, STD2 N2 2013 FUR2 1

The vertices in the network diagram below show the entrance to a wildlife park and six picnic areas in the park: **P1**, **P2**, **P3**, **P4**, **P5** and **P6**.

The numbers on the edges represent the lengths, in metres, of the roads joining these locations.



- a. In this graph, what is the degree of the vertex at the entrance to the wildlife park? (1 mark)
- b. What is the shortest distance, in metres, from the entrance to picnic area **P3**? (1 mark)

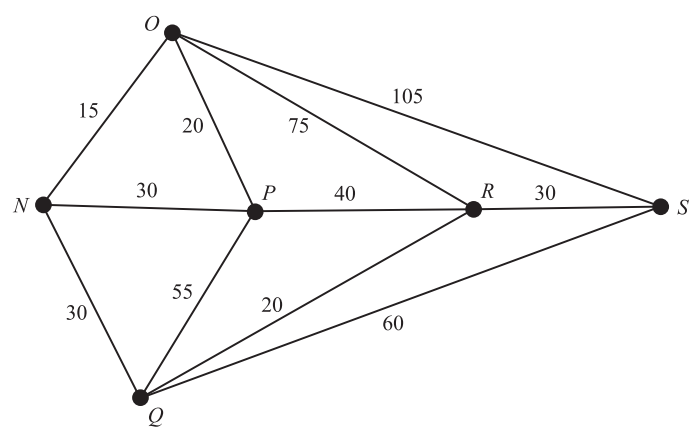
7. Networks, STD2 N2 2017 FUR2 1

Bus routes connect six towns.

The towns are Northend (**N**), Opera (**O**), Palmer (**P**), Quigley (**Q**), Rosebush (**R**) and Seatown (**S**).

The graph below gives the cost, in dollars, of bus travel along these routes.

Bai lives in Northend (**N**) and he will travel by bus to take a holiday in Seatown (**S**).

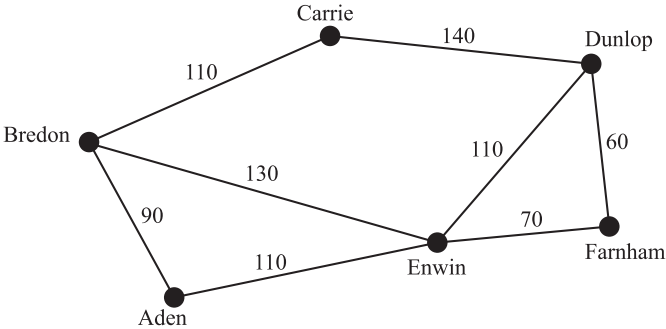


- a. Bai considers travelling by bus along the route Northend (**N**) – Opera (**O**) – Seatown (**S**). How much would Bai have to pay? **(1 mark)**
 - b. If Bai takes the cheapest route from Northend (**N**) to Seatown (**S**), which other town(s) will he pass through? **(1 mark)**
-

8. Networks, STD2 N2 2011 FUR2 1

Aden, Bredon, Carrie, Dunlop, Enwin and Farnham are six towns.

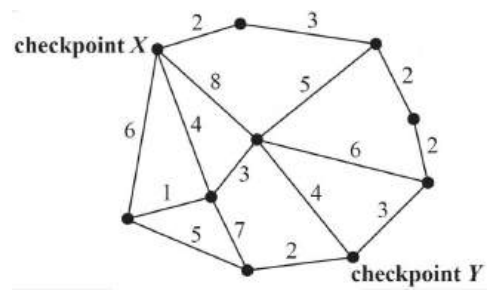
The network shows the road connections and distances between these towns in kilometres.



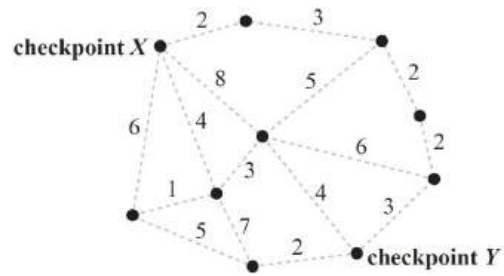
- a. In kilometers, what is the shortest distance between Farnham and Carrie? **(1 mark)**
 - b. How many different ways are there to travel from Farnham to Carrie without passing through any town more than once? **(1 mark)**
-

9. Networks, STD2 N2 2010 FUR2 2

The diagram below shows a network of tracks (represented by edges) between checkpoints (represented by vertices) in a short-distance running course. The numbers on the edges indicate the time, in minutes, a team would take to run along each track.

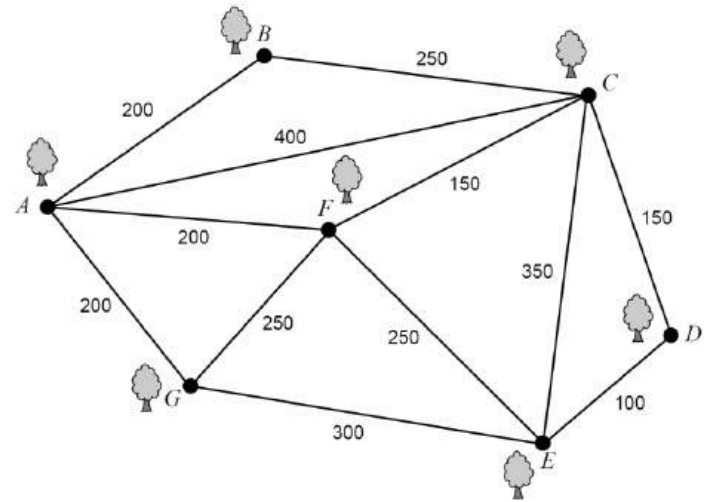


A challenge requires teams to run from checkpoint **X** to checkpoint **Y** using these tracks. What would be the shortest possible time for a team to run from checkpoint **X** to checkpoint **Y**? Mark the shortest route on the diagram below. (2 marks)



10. Networks, STD2 N2 SM-Bank 13

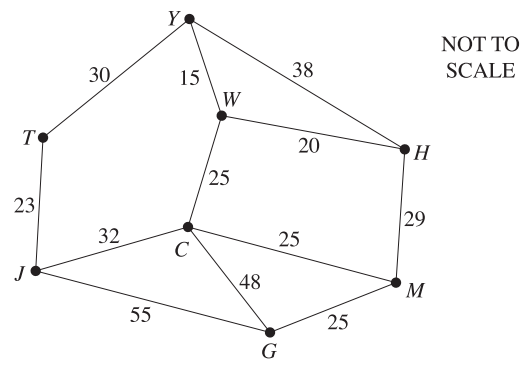
An estate has large open parklands that contain seven large trees. The trees are denoted as vertices **A** to **G** on the network diagram below. Walking paths link the trees as shown. The numbers on the edges represent the lengths of the paths in metres.



Jamie is standing at **A** and Michelle is standing at **D**. Write down the shortest route that Jamie can take and the distance travelled to meet Michelle at **D**. (1 mark)

11. Networks, STD2 N2 2024 HSC 16

A network of towns and the distances between them in kilometres is shown.



- a. What is the shortest path from **T** to **H** ? (2 marks)
- b. A truck driver needs to travel from **Y** to **G** but knows that the road from **C** to **G** is closed.
What is the length of the shortest path the truck driver can take from **Y** to **G** after the road closure? (2 marks)

12. Networks, STD2 N2 SM-Bank 28

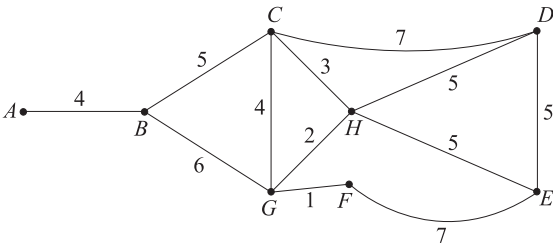
In central Queensland, there are four petrol stations **A**, **B**, **C** and **D**. The table shows the length, in kilometres, of roads connecting these petrol stations.

	A	B	C	D
A	—	170	—	150
B	170	—	160	90
C	—	160	—	120
D	150	90	120	—

- a. Construct a network diagram to represent the information in the table. (2 marks)
- b. A petrol tanker needs to refill each station. It starts at Station **A** and visits each station.
Calculate the shortest distance that can be travelled by the petrol tanker. In your answer, include the order the petrol stations are refilled. (2 marks)

13. Networks, STD2 N2 2019 HSC 30

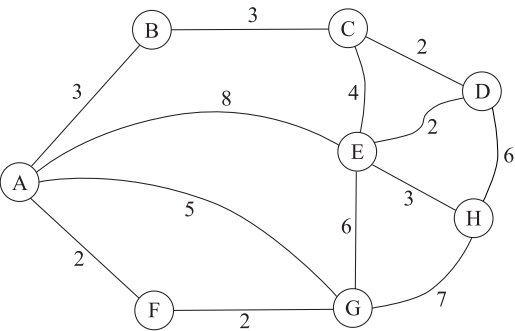
The network diagram shows the tracks connecting 8 picnic sites in a nature park. The vertices **A** to **H** represents the picnic sites. The weights on the edges represent the distance along the tracks between the picnic sites, in kilometres.



- a. Each picnic site needs to provide drinking water. The main water source is at site **A**.
Draw a minimum spanning tree and calculate the minimum length of water pipes required to supply water to all the sites if the water pipes can only be laid along the tracks. (2 marks)
- b. One day, the track between **C** and **H** is closed. State the vertices that identify the shortest path from **C** to **E** that avoids the closed track. (1 mark)

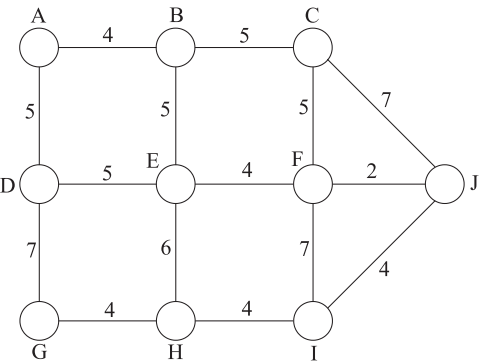
14. Networks, STD2 N2 SM-Bank 5

A network of roads is pictured below, with the distances of each road represented, in kilometres, on each edge.



- A driver wants to travel from A to H in the shortest distance possible.
- Describe the possible paths she can take, and the total distance she must travel. (2 marks)

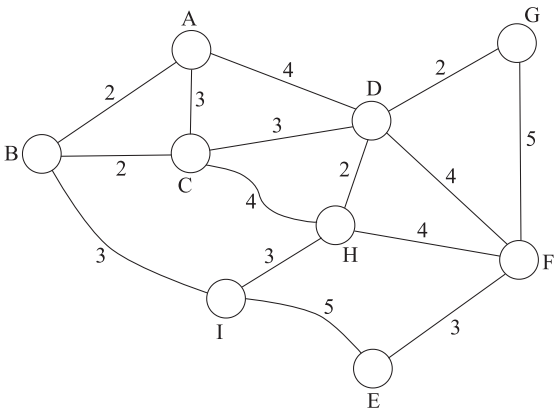
15. Networks, STD2 N2 SM-Bank 6



Describe the shortest path between A and J in the network above and its weight. (2 marks)

16. Networks, STD2 N2 SM-Bank 10

Consider the network pictured below.



Find the length of the shortest path from A to E. (2 marks)

17. Networks, STD2 N2 SM-Bank 12

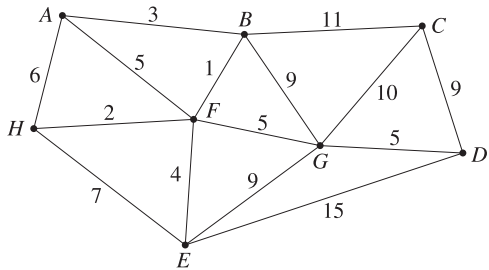
The following table shows the travelling time, in minutes, between towns which are directly connected by roads.

A dash indicates that towns are not directly connected.

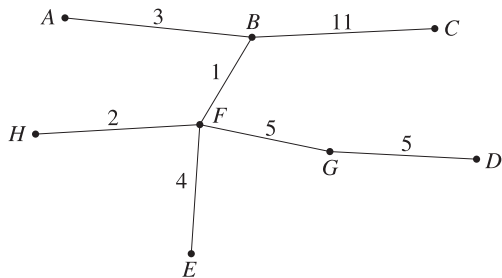
	A	B	C	D	E
A	0	50	18	10	—
B	50	0	28	32	20
C	18	28	0	—	50
D	10	32	—	0	80
E	—	20	50	80	0

- i. Draw a network diagram showing the information in this table. (2 marks)
- ii. What is the shortest travelling time between A and E? (1 mark)

A network of running tracks connects the points **A, B, C, D, E, F, G, H**, as shown. The number on each edge represents the time, in minutes, that a typical runner should take to run along each track.



- a. Which path could a typical runner take to run from point **A** to point **D** in the shortest time? (2 marks)
- b. A spanning tree of the network above is shown.



Is it a minimum spanning tree? Give a reason for your answer. (2 marks)

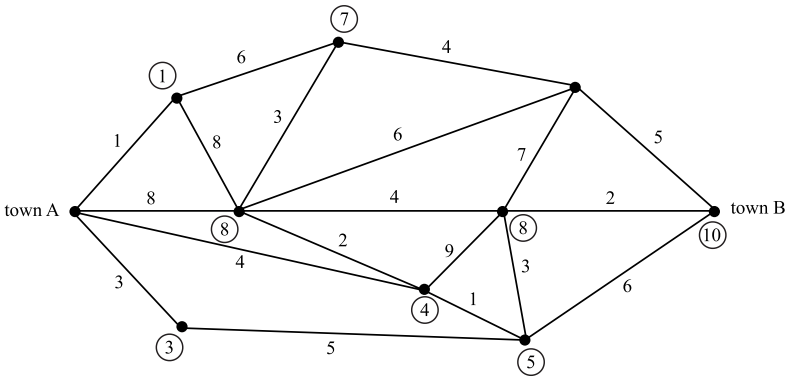
Worked Solutions

1. Networks, STD2 N2 2009 FUR1 2 MC

The shortest distance
= 2 + 4 + 1 + 1
= 8
⇒ **B**

2. Networks, STD2 N2 2007 FUR1 4 MC

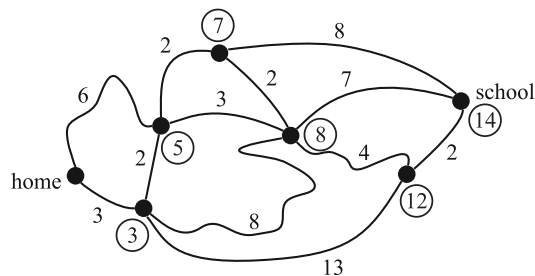
Using Dijkstra’s algorithm:



Shortest path
= 4 + 1 + 3 + 2
= 10
⇒ **B**

3. Networks, STD2 N2 2014 FUR1 3 MC

Using Dijkstra's algorithm:



Shortest time riding

$$= 3 + 2 + 3 + 4 + 2$$

$$= 14 \text{ minutes}$$

⇒ C

4. Networks, STD2 N2 SM-Bank 25 MC

Possible paths:

$$W-Z-X-Y = 120 + 70 + 140 = 330 \text{ m}$$

$$W-Z-Y-X = 120 + 100 + 140 = 360 \text{ m}$$

$$W-X-Z-Y = 150 + 70 + 100 = 320 \text{ m}$$

$$W-X-Y-Z = 150 + 140 + 100 = 390 \text{ m}$$

⇒ B

5. Networks, STD2 N2 2011 FUR2 1

a. Farnham to Carrie (shortest)

$$= 60 + 140$$

$$= 200 \text{ km}$$

b. Different paths are

FDC, FEDC, FEBC, FEABC, FDEBC, FDEABC

∴ 6 different ways

c. A possible path is *DFEABCDEB* and will finish

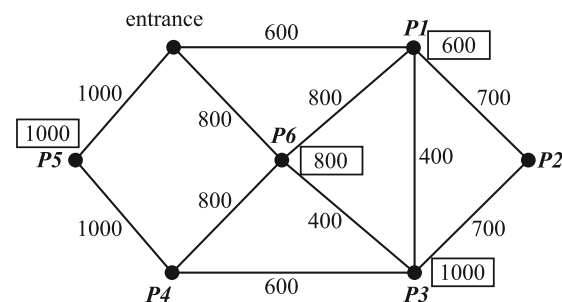
at Bredon – the only other odd-degree vertex.

(Note that solving this can be done quickly by applying the concept underlying the Konigsberg Bridge problem.)

6. Networks, STD2 N2 2013 FUR2 1

a. 3

b. Using Dijkstra's algorithm:



Shortest distance

$$= E-P1-P3$$

$$= 600 + 400$$

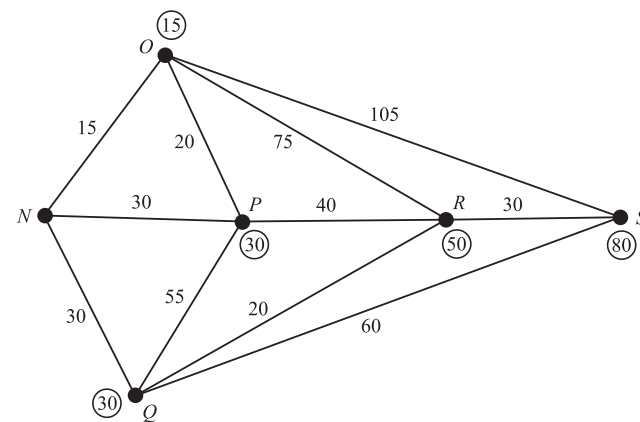
$$= 1000 \text{ m}$$

7. Networks, STD2 N2 2017 FUR2 1

a. Cost = 15 + 105

$$= \$120$$

b. Using Dijkstra's algorithm:



Fastest route is: *NQRS*.

∴ Other towns are Quigley and Rosebush.

8. Networks, STD2 N2 2011 FUR2 1

a. Farnham to Carrie (shortest)

$$= 60 + 140$$

$$= 200 \text{ km}$$

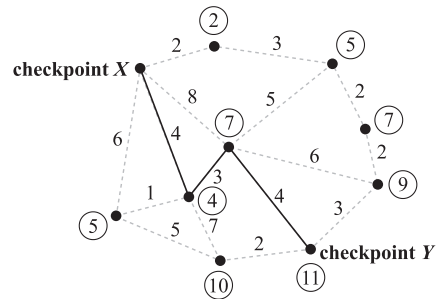
b. Different paths are

FDC, FEDC, FEBC,
FEABC, FDEBC,
FDEABC

$\therefore$ 6 different ways

9. Networks, STD2 N2 2010 FUR2 2

Using Dijkstra's Algorithm:



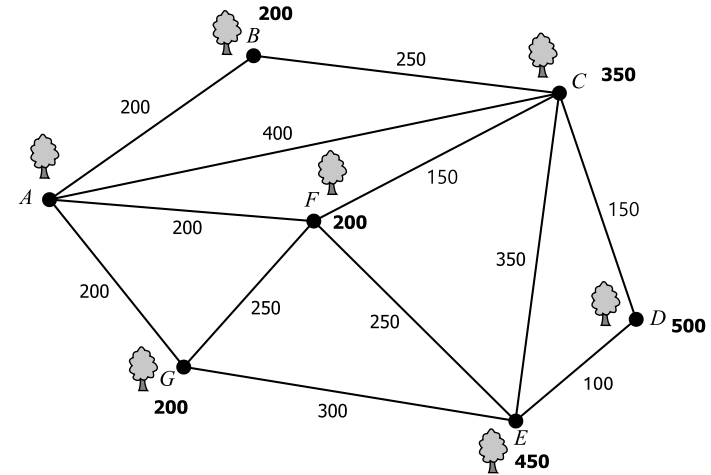
Shortest distance *X* to *Y*

$$= 4 + 3 + 4$$

$$= 11 \text{ minutes}$$

10. Networks, STD2 N2 SM-Bank 13

One strategy – using Dijkstra's algorithm:



Shortest path is *AFCD*

$$\text{Shortest distance} = 200 + 150 + 150$$

$$= 500 \text{ metres}$$

11. Networks, STD2 N2 2024 HSC 16

a. $TYH = 30 + 38 = 68$, $TYWH = 30 + 15 + 20 = 65$

$\therefore$ Shortest Path is *TYWH*.

b. $YWCMG = 15 + 25 + 25 + 25 = 90$

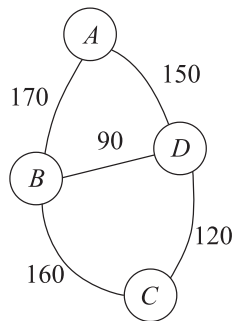
$$YWHMG = 15 + 20 + 29 + 25 = 89$$

$\Rightarrow$ All other paths are longer.

$\therefore$ Length of shortest path = 89 km

12. Networks, STD2 N2 SM-Bank 28

a.



b. Shortest Path from A (visiting all stations)

$A-B-D-C$

Distance = $170 + 90 + 120$

= 380 km

13. Networks, STD2 N2 2019 HSC 30

a. One strategy – using Prim's algorithm:

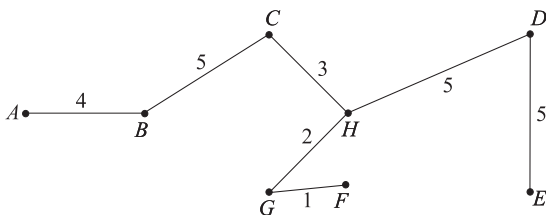
Starting at A

1st edge - AB, 2nd edge - BC

3rd edge - CH, 4th edge - HG

5th edge - GF, 6th edge - HD

7th edge - DE or HE

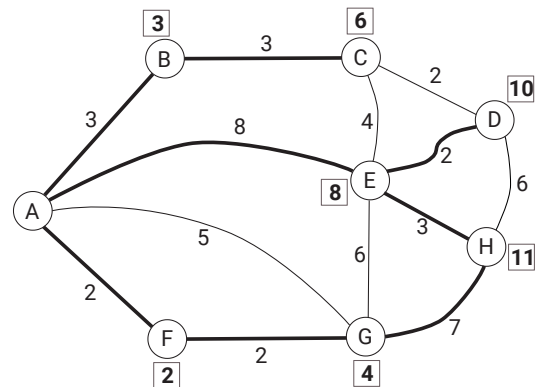


Maximum length = $4 + 5 + 3 + 2 + 1 + 5 + 5 = 25$ km

b. Shortest Path is CGHE

14. Networks, STD2 N2 SM-Bank 5

One Strategy: Dijkstra's algorithm



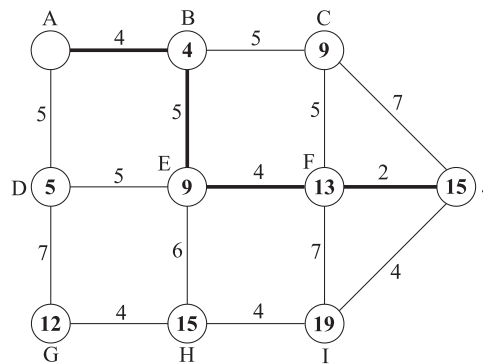
Shortest distance paths:

$A-E-H$ or $A-F-G-H$

Shortest distance is 11 km.

15. Networks, STD2 N2 SM-Bank 6

One Strategy: Using Dijkstra's algorithm



Shortest path is:

$A-B-E-F-J$

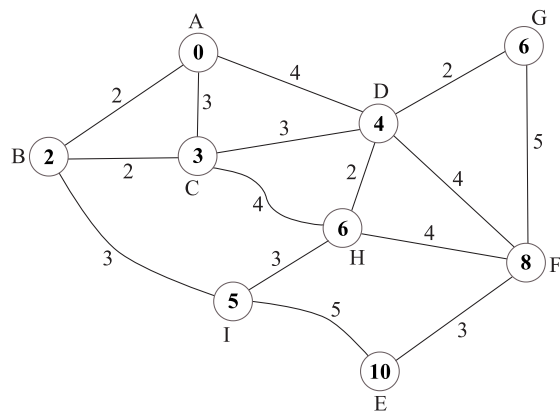
$\therefore$ Weight of shortest path

= $4 + 5 + 4 + 2$

= 15

16. Networks, STD2 N2 SM-Bank 10

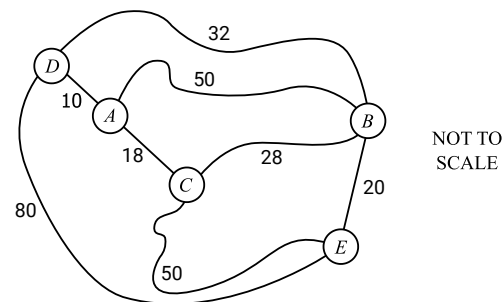
One strategy – Using Dijkstra's algorithm:



The shortest path A – B – I – E has a distance (weight) of 10.

17. Networks, STD2 N2 SM-Bank 12

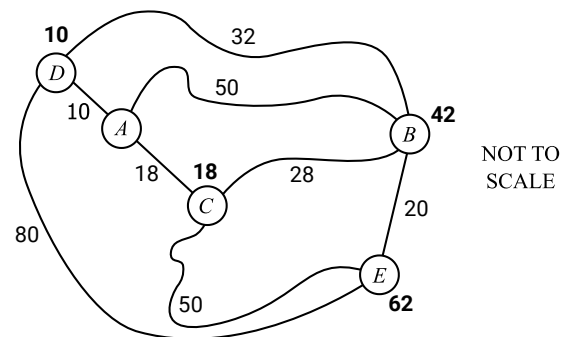
i.



Important to note that network diagrams do not need to be drawn to scale.

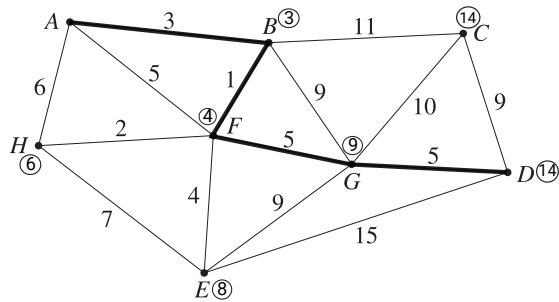
ii.

One strategy – using Dijkstra's algorithm:



∴ Shortest travelling time is the path A–D–B–E
 $= 10 + 32 + 20$
 $= 62 \text{ minutes}$

a. Using Dijkstra's Algorithm:



Shortest route = *ABFGD*

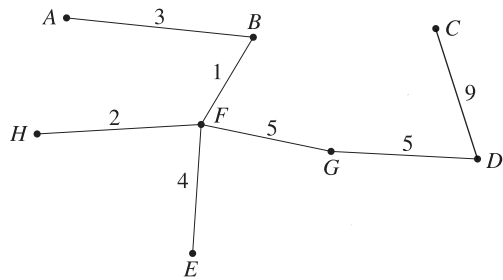
$$\begin{aligned} &= 3 + 1 + 5 + 5 \\ &= 14 \end{aligned}$$

b. Total time of given spanning tree

$$\begin{aligned} &= 3 + 11 + 1 + 2 + 4 + 5 + 5 \\ &= 31 \end{aligned}$$

♦ Mean mark (b) 47%.

Consider the MST below:



Total time (MST) = 3 + 1 + 2 + 4 + 5 + 5 + 9 = 29
∴ Given tree is NOT a MST.